Information Security

SPRING 2023

Vernam Cipher

- Exclusive OR- Operation
- Vernam-cipher, also called One-time pad (OTP) or the perfect cipher, is a crypto algorithm where plaintext is combined with a random key.
- The key is at least as long as the message or data that must be encrypted.
- Each key is used only once, and both sender and receiver must destroy their key after use.
- There should only be two copies of the key: one for the sender and one for the receiver.

Vernam Cipher

Rules of encryption is as follows:

- **Step-1:** Assign a number to each character like (a = 0, b = 1, c = 2, … z = 25). As per given table.

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Vernam Cipher

Rules of encryption is as follows:

- **Step-2:** Assign a number to each character of plain text and key according to alphabetical order. (From table of step-1).

Plain Text	I	A	M	C	H	I	R	A	G
	8	0	12	2	7	8	17	0	6
Key	**T**	**E**	**N**	**D**	**U**	**L**	**K**	**A**	**R**
	19	4	13	3	20	11	10	0	17

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Vernam Cipher

Rules of encryption is as follows:

- **Step-3:** Add both the number (Corresponding plain-text character number and Key character number).

Plain Text	I	A	M	C	H	I	R	A	G
	8	0	12	2	7	8	17	0	6
Key	**T**	**E**	**N**	**D**	**U**	**L**	**K**	**A**	**R**
	19	4	13	3	20	11	10	0	17
SUM (P.T. + Key)	27	4	25	5	27	19	27	0	23

Vernam Cipher

Rules of encryption is as follows:

- **Step-4:** Subtract the number from 26 if the added number is greater than 26. otherwise left it. Assign alphabets of numbers, it produces cipher text.

Plain Text	I	A	M	C	H	I	R	A	G
	8	0	12	2	7	8	17	0	6
Key	**T**	**E**	**N**	**D**	**U**	**L**	**K**	**A**	**R**
	19	4	13	3	20	11	10	0	17
SUM (P.T. + Key)	27	4	25	5	27	19	27	0	23
SUM – 26 (if SUM > 25)	1	4	25	5	1	19	1	0	23
Cipher Text	B	E	Z	F	B	T	B	A	X

Cipher Text: BEZFBTBAX

Vernam Cipher (Decryption Process)

Rules of decryption is as follows:

- **Step-1:** Assign a number to each character like (a = 0, b = 1, c = 2, … z = 25). As per given table.

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Vernam Cipher

Rules of decryption is as follows:

- **Step-2:** Assign a number to each character of cipher text and key according to alphabetical order.

Cipher Text	I	N	F	X	I	X	L	A	Y	N	L	H
	8	13	5	23	8	23	11	0	19	13	11	7
Key	A	N	T	I	I	N	D	I	A	N	Y	Z
	0	13	19	8	8	13	3	8	0	13	24	25

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Vernam Cipher

Rules of decryption is as follows:

- **Step-3:** Subtract cipher text alphabets number from key alphabet number. (Reverse process of encryption).

Cipher Text	**I**	**N**	**F**	**X**	**I**	**X**	**L**	**A**	**Y**	**N**	**L**	**H**
	8	13	5	23	8	23	11	0	19	13	11	7
Key	**A**	**N**	**T**	**I**	**I**	**N**	**D**	**I**	**A**	**N**	**Y**	**Z**
	0	13	19	8	8	13	3	8	0	13	24	25
SUM (C.T. - KEY)	8	0	-14	15	0	10	8	-8	19	0	-13	-18

Vernam Cipher

Rules of decryption is as follows:

- **Step-4:** If any number less than zero then add 26 in that number. Othe rw ise left it.

Cipher Text	**I**	**N**	**F**	**X**	**I**	**X**	**L**	**A**	**Y**	**N**	**L**
	8	13	5	23	8	23	11	0	19	13	11
Key	**A**	**N**	**T**	**I**	**I**	**N**	**D**	**I**	**A**	**N**	**Y**
	0	13	19	8	8	13	3	8	0	13	24
SUM (C.T. - KEY)	8	0	-14	15	0	10	8	-8	19	0	-13
SUM + 26 (if SUM < 0)	8	0	12	15	0	10	8	18	19	0	13
Plain Text	**I**	**A**	**M**	**P**	**A**	**K**	**I**	**S**	**T**	**A**	**N**

Class Task

- **Question:** Find out cipher text of below plain text using "Vernam Cipher".

Plain Text	**D**	**O**	**N**	**O**	**T**	**W**	**A**	**S**	**T**	**E**	**Y**	**O**	**U**	**R**	**T**	**I**	**M**	**E**
Keyword	L	I	F	E	I	S	S	U	S	P	I	C	I	O	U	S	D	O

Summary of today's lecture

- We discussed substitution cipher vernam cipher encryption decryption process.